Ravidu Rammuni Silva

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Summary

Doctoral researcher in Artificial Intelligence and Hourly Paid Lecturer in Computer Science at Nottingham Trent University, with over two years of commercial experience designing, building, and deploying production AI and embedded software for international clients in healthcare, manufacturing, and property technology. Research focuses on compact LLMs, post-training (SFT, LoRA/QLoRA, GRPO reinforcement learning) and execution-grounded inference for personalised education, alongside explainable AI (XAI) and computer vision, with a growing focus on verifiable AI for edge hardware. Work is published in *IEEE Transactions on Multimedia*, the *NeurIPS 2026 Workshop*, *UKCI*, and *SN Computer Science*, with open-source toolkits and benchmarks. Currently teaching C++ Programming, Mathematics for Computing, Database Engineering, and Software Design.

Education

Nottingham Trent University **10/2024 - Present**
PhD in Computer Science *Nottingham, UK*
- Theme: Leveraging Generative AI to enhance educational systems and deliver a personalised educational experience for transitional learners
- Expected completion: 2027

Nottingham Trent University **09/2022 - 09/2024**
MSc in Data Science (with integrated industrial placement) *Nottingham, UK*
- Award: Higher Distinction | GPA: 15.0 out of 16.0
- Dissertation: ArtBrain: Explainable AI for Classification and Attribution of AI-Generated Art and Style

University of Westminster **09/2016 - 09/2021**
BSc (Hons.) in Computer Science *London, UK*
- Award: First-Class Honours | Score: 81%
- Dissertation: Parallelised Deep Convolutional Neural Networks for Pathology Detection and Localisation in Chest X-Rays

University of Kelaniya **04/2018 - 04/2021**
BSc Physics, Pure Mathematics and Electronics *Kelaniya, Sri Lanka*
- Award: Second-Class Honours | GPA: 3.21 out of 4.00
- Project: Facial Recognition based Door-lock System

Teaching Experience

Nottingham Trent University, Department of Computer Science **08/2025 - Present**
Hourly Paid Lecturer *Nottingham, UK*

Contributed to the following modules alongside doctoral research:

- **Year 1 UG**: C++ Programming, Mathematics (for Computing)
- **Year 2 UG**: Database Engineering, Software Design and Implementation
- **BSc Apprenticeship**: Data Science Systems Analysis and Design

Research

Research interests: Generative AI and compact LLMs in education; execution-grounded reinforcement learning; explainable and verifiable AI (XAI); edge AI; computer vision and medical imaging.

Publications & Preprints

Visually Grounded Reinforcement Learning for LLM Programmatic Animation Generation in Interactive Execution Environments **12/2026**
Workshop | NeurIPS 2026 Workshop on Foundations of LLM Post-Training in Changing Environments | https://ravidusilva.com/assets/pdf/rammunisilva2026visually.pdf

ArtBrain: An Explainable End-to-End Toolkit for Classification and Attribution of AI-Generated Art and Style **10/2026**
Journal | IEEE Transactions on Multimedia | https://doi.org/10.1109/TMM.2026.3743638

LessonBench-V1: A Benchmark Dataset for Evaluating AI Lesson Generation Agents **07/2026**
Preprint | arXiv | https://doi.org/10.48550/arXiv.2607.13041

Training and Agentic Inference Strategies for LLM-based Manim Animation Generation **04/2026**
Preprint | arXiv | https://doi.org/10.48550/arXiv.2604.18364

Large Language Model Approaches to Educational Video Generation Using Manim **12/2025**
Conference | UKCI 2025 - Springer Nature | https://doi.org/10.1007/978-3-032-07938-1_26

Generative Artificial Intelligence and the Future of Education: A Systematic Review of Trends, Trustworthiness and Technological Evaluation **12/2025**
Preprint | Research Square | https://doi.org/10.21203/rs.3.rs-8248369/v1

FM-G-CAM: A Holistic Approach for Explainable AI in Computer Vision **12/2023**
Preprint | arXiv | https://doi.org/10.48550/arXiv.2312.05975

Recent Developments and Limitations in Recommender Systems: A Review **10/2022**
Conference | IEEE NILES 2022 | https://doi.org/10.1109/NILES56402.2022.9942405

Effective Utilization of Multiple Convolutional Neural Networks for Chest X-Ray Classification **09/2022**
Journal | SN Computer Science (Springer Nature) | https://doi.org/10.1007/s42979-022-01390-9

An Extensive Survey of Machine Learning Based Approaches on Automated Pathology Detection in Chest X-Rays **01/2021**
Conference | IEEE FRUCT 28 | https://doi.org/10.23919/FRUCT50888.2021.9347605

Full list & abstracts: https://ravidusilva.com/publications/ | **Google Scholar:** scholar.google.com/citations?user=IS2bTmwAAAAJ

Industry Experience

Ensera Design (formerly Kinneir Dufort) **09/2023 - 09/2024**
Software Engineer (Placement) *Bristol, UK*

Postgraduate placement within the MSc Data Science sandwich course, delivering software projects for multinational clients, primarily in healthcare.

- **AI-assisted breast cancer diagnosis** | *Python, PyTorch, FastAPI*: Developed the back end for an internal project deploying an existing CNN model, focusing on deployment architecture, application functionality and explainability of predictions.
- **AI natural stone classifier** | *C#, .NET, WinUI 3, C++*: Built a Windows application using the MVVM pattern for a Windows 11 IoT kiosk, integrating an SVM-based classifier; the application was validated and deployed to 80 kiosks.
- **Embedded drivers** | *C, C++, Python, CppUTest, STM32*: Developed a reusable CRC calculation module with unit tests for a generic driver library and a sound (beeper) driver for a medical device, plus a Python terminal application to automate release workflows.
- **Online catalogue** | *ReactJS, TypeScript, SharePoint*: Developed an internal management tool in SharePoint for a multinational healthcare company.

BooleenLab **07/2021 - 09/2022**
Engineer - AI Research & Technology *Colombo, Sri Lanka*

Promoted from Associate Engineer within six months. Led the Research and Development team on research-based commercial projects with clients in the USA, UK and Singapore.

- **Intelligent male STD support platform (HeHealth)** | *Python, PyTorch, Flask, AWS*: Led back-end and database development for a platform comprising a doctor portal, admin portal and mobile application; designed and deployed a two-fold, semi-serverless AWS architecture, and co-developed the CNN model, saliency maps and reasoning system.
- **Intelligent property inspection platform (Inspective)** | *Python, TensorFlow, Django, AWS*: Led back-end development of an admin panel and mobile application that lets tenants carry out property inspections, automatically identifying defects and generating housing-authority forms; deployed a custom CNN on AWS.
- **Recommender system for fabrics and elastics** | *Python, PyTorch, FastAPI, AWS*: Initiated and led research on fabric classification for Stretchline, towards an intelligent e-catalogue that recommends products based on customer preferences.

Rootcode Labs **05/2021 - 06/2021**
AI Engineer - Intern *Colombo, Sri Lanka*

- **ConverseAI recruitment chatbot** | *Python, Rasa, FastAPI, AWS*: Implemented a retraining pipeline with around twenty seconds of downtime; the product won an innovation award.
- **Intelligent CV parser** | *Python, PyTorch, NLTK, AWS*: Contributed to training and retraining pipelines for extracting data from professional CVs.

Subject Expertise & Technical Skills

Programming and Software Development: Python; C++; C#; embedded C (STM32); Java; Swift; JavaScript/TypeScript; MVVM; FastAPI, Django, Flask, ReactJS; Git, GitHub, Bitbucket, Jira

AI, Machine Learning & Data Science: PyTorch, TensorFlow, TensorFlow.js; CNNs, explainable AI (XAI), compact LLMs (SFT, LoRA/QLoRA, GRPO reinforcement learning, execution-grounded inference)

Cloud Computing: AWS, GCP, Azure, IBM Watson (design and deployment of production-grade architectures)

Mobile Application Development: iOS/Swift; Django-based back ends for production mobile applications

Databases: PostgreSQL, MySQL; database design for production back ends; currently teaching Database Engineering

Certifications: Deep Learning Specialisation (Coursera); Advanced Data Science Specialisation (IBM): 1, 2, 3, 4

Professional Activities

Nottingham Trent University **12/2025 - Present**
PGR Ambassador, Department of Computer Science *(Represents PGRs in NTU CIRC)*

Nottingham Trent University **11/2022 - Present**
Student Ambassador and Student Outreach Ambassador *(Promoted university education across Nottinghamshire schools)*

Peer Reviewing **02/2020 - Present**
IEEE FRUCT Conference, IEEE TMM Journal, IEEE IJCNN Conference, IEEE IROS Conference

AIESEC International **10/2019 - 06/2020**
Mobile Application Developer (Volunteer)

References

Available on request.